

Project Requirements Document:

Canadian Federal Procurement Analysis – Competition, Vendor Concentration, and Spending Trends

BI Analyst: Jatin Sagwal

Client/Sponsor: Self-directed portfolio project. Simulated sponsor persona: Director, Internal Audit & Procurement Oversight, Government of Canada (hypothetical, used to ground the requirements in a realistic organizational context)

Purpose: Federal procurement is one of the largest and most discretion-heavy categories of public spending in Canada, and it is a recognized area of fraud and inefficiency risk. Despite this, no consolidated, decision-ready dashboard currently exists that translates the raw proactive-disclosure contract files into clear insight on competition levels, vendor concentration, and spending patterns. This project is worth the investment of analyst time because it (1) produces a reusable transparency tool that oversight staff, journalists, or the public could act on, and (2) demonstrates, in a portfolio context, the practical data-cleaning, analytical, and visualization skills used in real government and corporate risk-analytics work.

Key dependencies:

- **Data source:** Open Government Portal – “Proactive Publication - Contracts” dataset (contracts.csv), published by Treasury Board of Canada Secretariat, updated periodically.
- **Tools:** Python (pandas) or SQL for data cleaning and vendor-name standardization; Power BI or Tableau for the interactive dashboard; GitHub for version control and portfolio hosting.
- **Team:** Solo analyst (this project), with informal peer/mentor review at the draft-findings stage.
- **Contact for data questions:** Open Government team, Treasury Board of Canada Secretariat (open-ouvert@tbs-sct.gc.ca) – no direct departmental access is required since the data is fully public.
- **Expected deliverables:** cleaned/documented dataset, interactive dashboard, written findings memo, and a public summary post (LinkedIn/GitHub README).

Stakeholder requirements: (from the Stakeholder Requirements

Document, prioritized as R – required, D – desired, N – nice to have)

Requirement	Priority
Filter by fiscal year, department, vendor, and contract type	R
Competitive vs. sole-source breakdown by department	R
Top vendors by contract value, with drill-down	R
Vendor concentration metric (HHI or top-10 share), government-wide and by department	R
Five-plus year trend view	R
Year-end (March) spending-spike flag	D
Refreshable as new disclosure files are published	D
Source/methodology note on all visuals	R

Success criteria: (SMART)

- Deliver a published, interactive dashboard covering at least five fiscal years of contract data within four weeks of project start.
- Identify and document at least three data-backed findings (e.g., government-wide sole-source rate, top-vendor concentration %, department-level comparison) each supported by a chart.
- Ensure the dashboard remains responsive when filtering the full multi-hundred-thousand-row dataset, using pre-aggregation where needed.
- Publish the findings write-up and a public summary within one week of dashboard completion.

User journeys:

- **Current:** Anyone wanting this insight – oversight staff, a journalist, or a member of the public – must manually download a large raw CSV from the Open Government Portal, clean inconsistent vendor names, filter by department/year, and calculate concentration metrics by hand in a spreadsheet. This is slow, error-prone, and inaccessible to non-technical users.
- **Future:** The user opens the dashboard, selects a fiscal year and department, and immediately sees the competition rate, top vendors, and concentration index for that selection, with the option to drill into individual contract records – without touching the raw data at all.

Assumptions:

- Data published by Treasury Board Secretariat is materially complete and representative, notwithstanding the source's "unaudited" disclaimer.

- The fiscal year field aligns with the Government of Canada fiscal year (April 1 – March 31), not the calendar year.
- Vendor name variants (e.g., abbreviations, punctuation differences) can be reasonably reconciled through fuzzy matching without an authoritative vendor master list.
- The analysis is retrospective and observational; without access to internal procurement justifications, “sole-source” will be treated as a risk indicator, not evidence of wrongdoing.

Compliance and privacy:

- All data used is published under the Open Government Licence – Canada and is explicitly intended for public reuse; no additional access approval is required.
- The dataset is business/vendor information; where individual public servants’ names appear (e.g., contracting officer fields), these will be excluded or anonymized from any public-facing dashboard or report to avoid unintended exposure of individuals.
- Findings will be framed as “patterns warranting further review” rather than claims of fraud or wrongdoing, consistent with responsible public-data analysis and journalism norms.

Accessibility:

- Use colour palettes with sufficient contrast; avoid encoding competitive vs. sole-source using colour alone (e.g., pair colour with icons or labels) for colour-blind users.
- All charts include descriptive titles and axis labels rather than relying on legends alone.
- Enable platform-native keyboard navigation and screen-reader support (Power BI/Tableau accessibility features).
- Accompany the dashboard with a plain-text findings memo so key results are available without relying on the visuals.

Roll-out plan: (four-week scope)

- **Week 1:** Data acquisition, field inventory, data cleaning, and vendor-name standardization.
- **Week 2:** Exploratory analysis and metric definition (competition rate, HHI, top-vendor share); draft findings.
- **Week 3:** Dashboard build in Power BI/Tableau, with iterative self-review.
- **Week 4:** Finalize findings memo, publish dashboard, and write the public summary/portfolio post.
- **Priority order:** competition/sole-source analysis first (core narrative), vendor concentration second, year-end spending-spike analysis third (stretch goal if time permits).